

Nifty 100 Analytics

Weekly Status Report — All Sprints (1-6)

Project: Fundamental analytics platform for the Nifty 100 index — ETL, ratio engine, screener, peer comparison, Streamlit dashboard, PDF reports, NLP pros/cons, KMeans clustering, and a REST API.

Duration: 45 working days across 6 one-week sprints.

Project Timeline — At a Glance

Week	Sprint / Focus	Days	Story Points	Status
Week 1	Sprint 1 — Data Foundation	Day 01-07	34 SP	Complete
Week 2	Sprint 2 — Financial Ratio Engine	Day 08-14	42 SP	Complete
Week 3	Sprint 3 — Screener + Peer Engine	Day 15-21	49 SP	Complete
Week 4	Sprint 4 — Dashboard + Valuation	Day 22-28	55 SP	Complete
Week 5	Sprint 5 — Cash Flow Intel + Reports + NLP	Day 29-35	70 SP	Complete
Week 6	Sprint 6 — Clustering + API + Sign-off	Day 36-45	89 SP	Complete

Week 1 — Data Foundation

Load and validate all source data into a single SQLite database.

Status: Complete

What shipped

- 10-table SQLite schema (companies, P&L, balance sheet, cash flow, analysis, documents, pros/cons, sectors, stock prices, peer groups).
- ETL loader + normaliser handling 5 different raw date/ticker formats found in the real files.
- All 16 data-quality rules implemented; CRITICAL failures excluded from the load and logged.
- 46 automated tests (20 year-parsing + 15 ticker + 10 loader + 1 extra), 0 failures.

Result

nifty100.db fully populated and validated — the foundation every later sprint reads from.

Week 2 — Financial Ratio Engine

Compute 50+ financial KPIs for every company-year, with edge cases handled correctly.

Status: Complete

What shipped

- Profitability, leverage and efficiency ratios: NPM, OPM (cross-checked), ROE, ROCE, ROA, D/E, ICR, Net Debt, Asset Turnover.
- CAGR engine (Revenue/PAT/EPS, 3/5/10-year) with all 6 edge cases: normal, turnaround, decline-to-loss, both-negative, zero-base, insufficient-data.
- Cash-flow KPIs: FCF, CFO Quality Score, CapEx Intensity, FCF Conversion, 8-pattern capital allocation classifier.
- financial_ratios table: 1,164 rows, 46 columns (exceeds the 1,100-row target).
- 42 unit tests, 0 failures (exceeds the 20-test target).

Week 3 — Screener + Peer Engine

A working stock screener with 6 presets, plus peer percentile rankings.

Status: Complete

What shipped

- Analyst-editable screener config (15 filterable metrics) and a generic threshold filter engine.
- 6 presets (Quality Compounder, Value Pick, Growth Accelerator, Dividend Champion, Debt-Free Blue Chip, Turnaround Watch), all tested against the full 92-company universe.
- Sector-relative, winsorized composite quality score (v2).
- Peer percentile engine: 10 metrics across 11 peer groups; 92 radar-chart PNGs.
- screener_output.xlsx (6 sheets) and peer_comparison.xlsx (11 sheets), both colour-coded.

Week 4 — Streamlit Dashboard + Valuation

An 8-screen interactive dashboard plus a valuation flagging module.

Status: Complete

What shipped

- 8 dashboard screens: Home, Company Profile, Screener, Peer Comparison, Trend Analysis, Sector Analysis, Capital Allocation Map, Annual Reports.
- Cached data-loading layer shared by every screen (@st.cache_data, 10-min TTL).
- Valuation module: FCF yield, sector-median P/E, Fair / Caution / Discount flags for all 92 companies.
- CSV export from the Screener screen; live annual-report link validity checks.

Week 5 — Cash Flow Intelligence + Reports + NLP

Auto-generated pros/cons, deeper cash-flow classification, and PDF reporting.

Status: Complete

What shipped

- NLP parser for free-text growth fields + 12 pro-rules / 12 con-rules pros-cons generator (549 rows, every company covered).
- Cash flow intelligence: distress-signal and deleveraging flags on top of the Sprint 2 cash-flow KPIs.
- 91 of 92 two-page company tearsheet PDFs (1 skipped — insufficient history, documented).
- 10 sector PDF reports and a portfolio summary PDF.

Week 6 — Clustering + REST API + QA + Sign-off

KMeans company archetypes, a full REST API, and final acceptance sign-off.

Status: Complete — 20/20 gates fully pass

What shipped

- KMeans clustering (k=5): all 92 companies assigned an archetype, elbow plot + correlation heatmap + outlier report.
- FastAPI REST service: 16 endpoints (companies, screener, sectors, peers, valuation, portfolio, health).
- docs/openapi.json + Postman collection + a 10+ page analyst guide PDF.
- Final acceptance run against 20 gates: 15 full PASS, 3 PARTIAL (data-quality notes, not defects), 2 NOT VERIFIED pending a live server run with network access.

Open items for next week

- Run `make api` on a machine with network access and re-check gates AC-11 (health endpoint) and AC-13 (API vs. screener match).
- Re-confirm AC-08 (Company Profile screen load time) with a live Streamlit session.